

NETWORK LAYER

Module 3

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Solution: Do it in the class.

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Answer: Range of usable IP address: 192.168.17.1/25 – 192.168.17.126/25

Example: An organization has been allocated an IPv4 address block 210.50.0.0/16 (a total of 65,536 addresses). The network administrator must divide this block among three departments with the following requirements:

- Department A: 50 subnets, each requiring 300 usable addresses
- Department B: 100 subnets, each requiring 150 usable addresses
- Department C: 120 subnets, each requiring 60 usable addresses

Design the subnet blocks for each department using VLSM, determine the total number of allocated addresses, and compute how many addresses remain unassigned after these allocations.

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Solution:

Department A: 210.50.0.0 – 210.50.99.255, using $50 \times /23$

Department B: 210.50.100.0 – 210.50.199.255, using $100 \times /24$

Department C: 210.50.200.0 – 210.50.229.255, using $120 \times /26$

Total allocated: 58,880 addresses

Remaining: 6,656 addresses

Example: A company has three branch offices Central, East, and West connected through private WAN links. The organization has been assigned an IPv4 block of 70.12.100.128/26, which provides 64 addresses. Management decides to allocate 32 addresses to the Central office, while the remaining addresses must be divided between the East and West offices. Design the subnets for all three offices and specify the subnet address, first usable address, last usable address, and broadcast address for each location.

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Solution:

Office	Subnet	Subnet Address	First Usable	Last Usable	Broadcast
Central	/27	70.12.100.128	70.12.100.129	70.12.100.158	70.12.100.159
East	/28	70.12.100.160	70.12.100.161	70.12.100.174	70.12.100.175
West	/28	70.12.100.176	70.12.100.177	70.12.100.190	70.12.100.191

References:

Data Communications and Networking by Behrouz A. Forouzan.



THANK YOU